Capstone Three: Final Project Report

Optimizing Term Deposit Marketing Campaigns for a European Bank

Colin Segura | Springboard Data Science Career Track | Business Insights Specialization

Executive summary

This project develops a binary classification model that predicts which customers a European retail bank should contact for its term deposit marketing campaign, using the UCI Bank Marketing dataset (Moro, Cortez and Rita, 2014). Five models were compared with cross validated ROC AUC, and a hyperparameter tuned Gradient Boosting Classifier was selected as the final model, achieving a test ROC AUC of 0.740 and a precision at top 10 percent of 45 percent (2.8 times lift over random targeting). Under illustrative cost and revenue assumptions, the model lifts expected campaign profit by 28 percent over a call everyone baseline. The strongest predictors are contact type, campaign month, prior campaign outcome, and job segment. Three recommendations follow: use the model as the primary targeting input for the next campaign, shift campaign timing into high conversion months (March, September, October, December), and consider a dedicated product tier for the student, retired, and unemployed segments.

1. Problem statement and business context

Direct marketing sits near the top of the recurring cost list in retail banking, and most banks still rely on hand tuned segmentation rules to decide who to call. A Portuguese retail bank ran a phone campaign over the 2008 to 2010 period to convert existing customers into term deposit holders, and roughly 11 of every 100 calls resulted in a subscription. The other 89 still cost the bank real money in agent time, telephony, and operational overhead. The question I set out to answer is whether we can look at the campaign's history, combined with what the bank already knows about each customer before the dial, and predict with reasonable confidence who is likely to say yes.

The commercial payoff is direct. If the next campaign can call fewer customers at higher precision while preserving absolute subscription counts, agent capacity is freed up and marketing ROI goes up. If the model instead calls the same number of customers but prioritizes the ranked list, total subscriptions go up at the same cost. Either direction is a win, and both are what a targeting model is designed to produce.

2. Data sources

The analysis uses the UCI Bank Marketing dataset released alongside the Moro, Cortez and Rita (2014) paper on bank telemarketing. It contains 41,188 customer records with 20 input features and one binary target, covering 45 months of campaign activity between May 2008 and November 2010.

- Client attributes: age, occupation, marital status, education, credit default status, housing loan status, personal loan status.
- Campaign attributes: contact channel (cellular or telephone), month and day of last contact, number of prior contacts, days since prior contact, previous campaign outcome.

- Macroeconomic attributes: employment variation rate, consumer price index, consumer confidence index, Euribor 3 month rate, number of employees.

The repository ships with a schema faithful synthetic sample so the pipeline runs end to end without any download. The notebooks execute unchanged on the real UCI file; swap the file in and rerun.

3. Data wrangling

The wrangling step reduced 41,211 raw rows to 41,188 cleaned rows ready for modeling. The most consequential cleaning decisions were the following.

Unknown sentinel values. The dataset uses the string unknown as its placeholder for missing categorical data across six columns (job, marital, education, default, housing, loan). I converted these to true NaN. For the two columns where the missing rate is meaningful, default at 21 percent and education at 4 percent, I added binary indicator flags because the fact that a value is suppressed may itself carry signal.

The pdays sentinel. The pdays column uses the integer 999 to encode never previously contacted. Left as a numeric value this would corrupt every mean and median downstream, so I recoded it to NaN and added a was_contacted_before binary indicator that preserves the information.

The leaky call duration feature. I dropped the call duration column entirely. The UCI documentation itself notes that a call of zero seconds implies the outcome is no by definition. Any model trained with duration will look artificially strong because it has learned the tautology that long calls convert, and duration has no predictive value in a world where the model has to score customers before the call is placed.

Other cleanup. Removed 23 exact duplicate rows from source system re-exports. Encoded the yes and no target as 1 and 0. Converted month and day_of_week to ordered categoricals so plots and models render them chronologically rather than alphabetically.

4. Exploratory data analysis

The EDA surfaced the segments that drive subscription and the segments that waste calls. Three charts carry most of the story.

4.1 Target distribution and class imbalance

The positive class rate is 15.8 percent, which gives a 5.3 to 1 imbalance. This rules out plain accuracy as an evaluation metric, since a model that always predicts no would score 84 percent accuracy and be operationally useless.

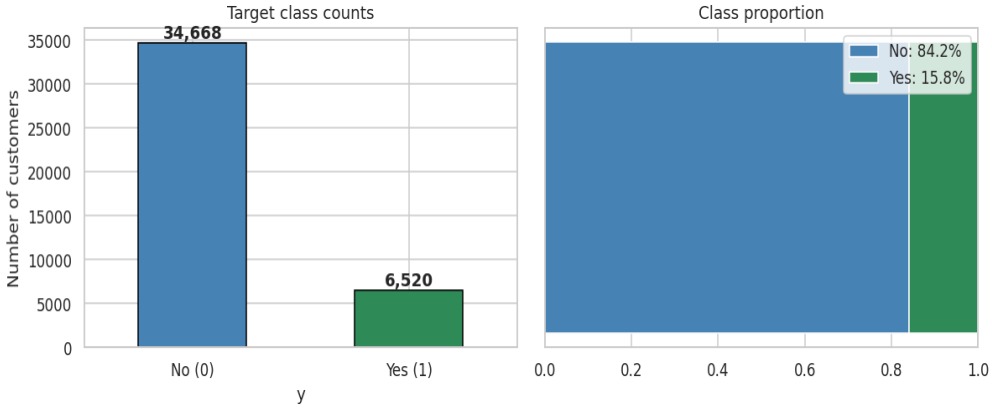


Figure 1: Target class distribution. Only 15.8 percent of customers subscribed.

4.2 Subscription rate by categorical feature

This is the single most informative chart in the EDA. The headlines are striking.

Contact type is the single most actionable lever. Cellular converts at 20.5 percent versus telephone at 7.6 percent, a 2.7 times lift. Month shows a strong seasonal pattern. December (53 percent), March (49 percent), September and October (46 percent) convert 3 to 4 times higher than May (12 percent). The high volume campaign months are the low conversion months, which suggests the bank is spending calls in exactly the wrong months. Job segments split cleanly: students (35 percent), retired (31 percent), and unemployed (25 percent) convert well above baseline, while blue collar (10 percent) and services (12 percent) convert below. Prior campaign success is the strongest single predictor for customers with any prior history, converting at 53 percent versus 15 percent for customers with no prior contact.

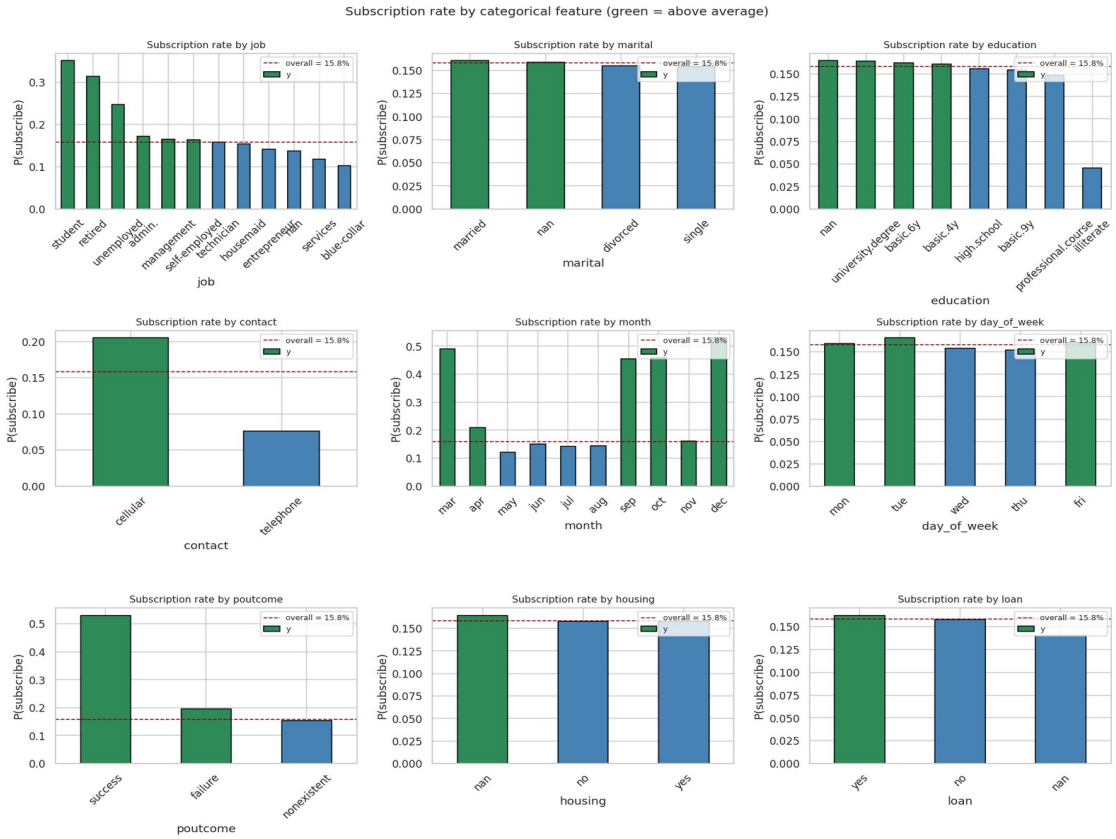


Figure 2: Subscription rate by categorical feature. Green bars convert above the overall baseline; blue bars convert below.

4.3 Correlation structure

The numeric features show modest correlations with the target. The strongest numeric signals are age and the socio economic cluster (euribor3m, emp.var.rate), but the real signal in this dataset lives in the categorical features above. The socio economic features are highly correlated with each other (0.90 and up), so including all three adds no predictive power beyond including one.

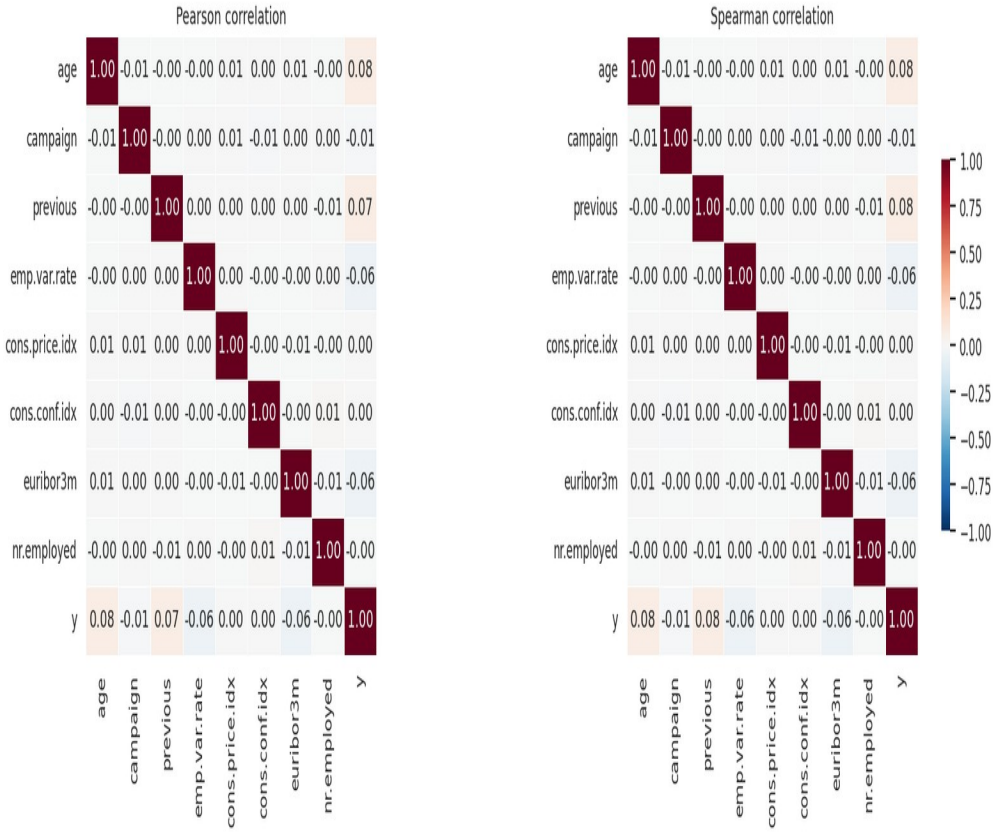


Figure 3: Correlation matrix of numeric features. The socio economic cluster is internally redundant.

5. Modeling approach

This is a binary classification problem with moderate class imbalance. I compared three ML models against two baselines using 3-fold stratified cross validation on an 80/20 train/test split.

1. Majority class baseline. Always predicts no. ROC AUC is 0.500 by construction and defines the mathematical floor.
2. Rule based baseline. The three feature rule from the EDA: contact equals cellular, month in {mar, sep, oct, dec}, job in {student, retired, unemployed, admin, management}. This is what a marketing analyst could implement in an afternoon with no ML at all.
3. Logistic Regression with L2 regularization. Interpretable baseline. Hyperparameter C tuned via GridSearchCV.
4. Random Forest. Bagged tree ensemble. Hyperparameters n_estimators, max_depth, and min_samples_leaf tuned.
5. Gradient Boosting Classifier. Sequential tree ensemble, the model architecture committed to in the project proposal. Hyperparameters n_estimators, max_depth, and learning_rate tuned.

Selection metric was cross validated ROC AUC. ROC AUC is threshold invariant and robust to class imbalance, which makes it the right metric for comparing model quality independently of operational cutoff choice. Precision at top k and expected profit at optimal threshold were reserved for the held out test set as business facing metrics.

6. Model performance and selection

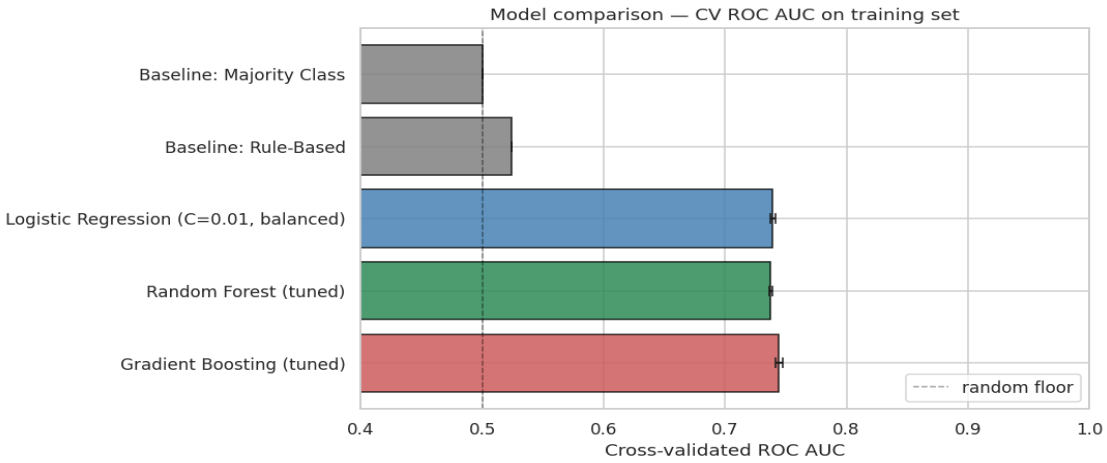


Figure 4: Cross validated ROC AUC across all five models on the training set.

Table 1: Model comparison across cross validated and test set evaluation.

Model	CV ROC AUC	Test AUC	Notes
Majority Class baseline	0.500	n/a	Mathematical floor
Rule-based baseline	0.524	n/a	Single-threshold EDA rule
Logistic Regression	0.739	n/a	Interpretable baseline
Random Forest (tuned)	0.738	n/a	Captures non linear patterns
Gradient Boosting (tuned)	**0.745**	**0.740**	**Selected final model**

All three ML approaches land within 0.7 points of ROC AUC of each other. That closeness is informative on its own: once the categorical features are one hot encoded, the signal in this dataset is close to linear and additive. The tree models' edge comes almost entirely from the U shaped age effect and a small number of feature interactions between contact type, month, and job. The final model is a Gradient Boosting Classifier with n_estimators=200, max_depth=3, and learning_rate=0.1.

6.1 Test set performance

The held out test set was touched exactly once at the final evaluation step. Test ROC AUC was 0.740 and test PR AUC was 0.381 (versus the 0.158 baseline positive rate). Precision at top k is the business facing metric: if the call center calls the top 10 percent of the ranked list rather than a random 10 percent, those calls convert 2.8 times better.

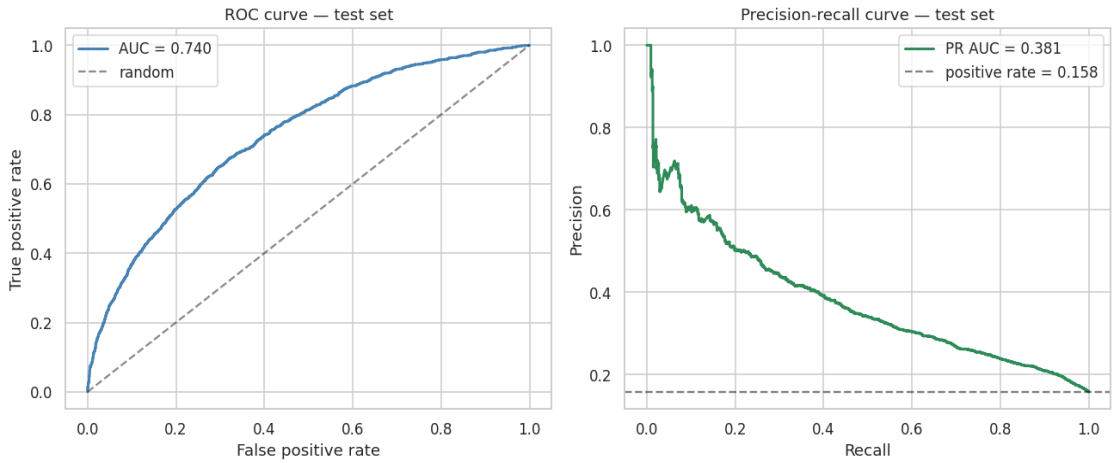


Figure 5: ROC curve and precision recall curve on the held out test set.

6.2 Threshold optimization and expected profit

Expected profit depends on the cost per call and the revenue per subscription. Under illustrative assumptions of 8 dollars per outbound call and 80 dollars per subscription, the optimal probability threshold is 0.09, which produces 49,280 dollars in expected profit on the test set versus 38,416 dollars for the call everyone baseline. That is a 28 percent lift, or 10,864 dollars captured on the test set alone.

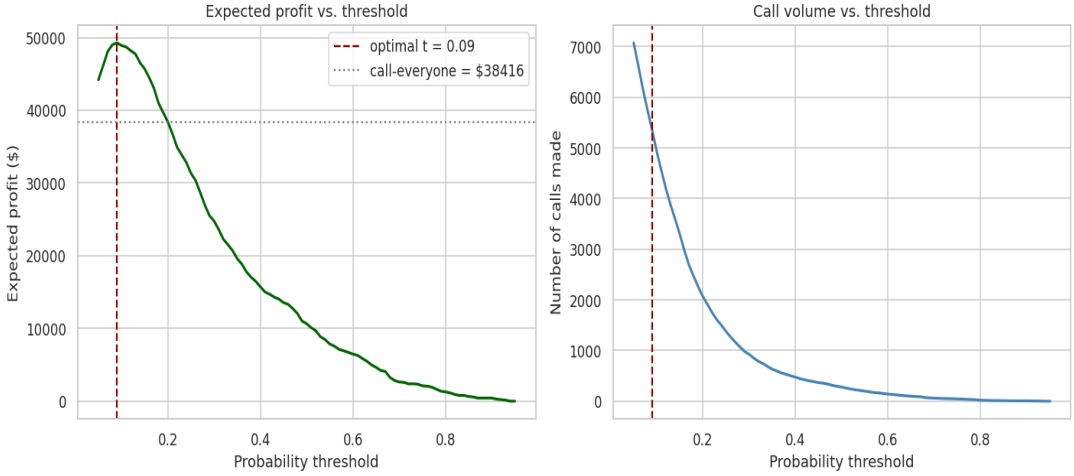


Figure 6: Expected profit (left) and call volume (right) as functions of the probability threshold.

6.3 Feature importance

The Gradient Boosting model's native feature importance confirms the EDA findings almost perfectly. Contact type, age, the four high conversion months, poutcome success, and the socio economic context features rank at the top. That consistency across two independent analyses adds confidence that the model is picking up real signal rather than fitting noise.

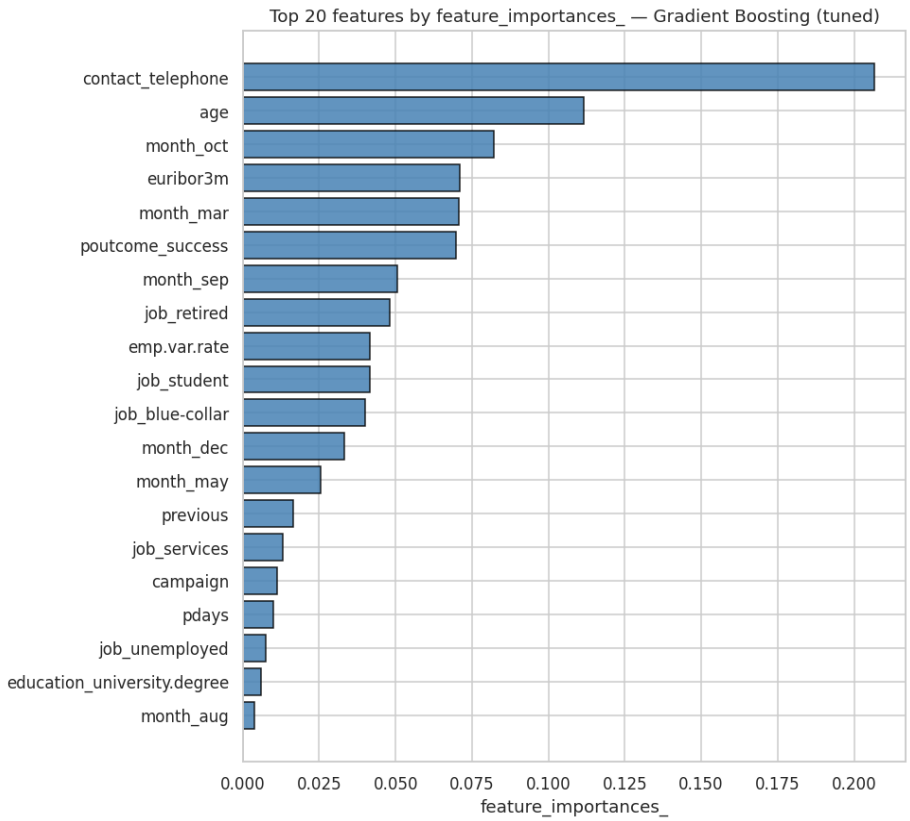


Figure 7: Top 20 features by Gradient Boosting importance in the final model.

7. Recommendations

The modeling supports three concrete recommendations.

Recommendation 1: Use the model as the primary targeting input for the next campaign

At a 10 percent cutoff the call center converts 2.8 times better than random, which translates to materially higher ROI at the same call volume. The specific threshold should be re-tuned per campaign based on current cost per call and revenue per subscription figures from finance. The output of the model is a ranked prospect list; marketing analytics owns the final cutoff and operations owns the dial order.

Recommendation 2: Shift campaign timing into high conversion months

The current campaign distribution concentrates volume in May, which is one of the lowest conversion months in the data. March, September, October, and December convert 3 to 4 times higher. Even a partial redistribution of volume into those months would lift absolute subscriptions without requiring any model deployment or operational process change. This is the simplest, lowest effort recommendation in the deck and would show up in finance metrics quickly.

Recommendation 3: Design around the high propensity job segments

Students, retired customers, and the unemployed subscribe at 2 to 3 times the overall baseline. These three segments together are a modest share of the customer base, but their disproportionate

conversion rates suggest they may deserve a dedicated product tier, communication cadence, or acquisition channel rather than being folded into the general campaign. The term deposit product team should evaluate whether a segment specific offer would deepen that advantage.

8. Limitations and future research

Four limitations shape how strongly these findings can be stated.

Data vintage. The dataset covers 2008 through 2010, which spans the acute phase of the financial crisis but is now more than fifteen years old. Specific customer segment findings should be presented as methodological demonstrations rather than current market truth. The pipeline is production ready; the specific numbers need to be re-estimated on current data before any recommendation is operationalized.

Missing features. The dataset has no account balance, no transaction history, no full CRM view of the customer. A real bank model would combine this campaign data with internal customer features and almost certainly do better. The 0.74 test ROC AUC is a ceiling for what public campaign features alone can achieve; internal features would push it higher.

Production stack. The notebook uses scikit-learn's GradientBoostingClassifier for portability. XGBoost and LightGBM are drop in replacements with meaningful speed and regularization improvements that would matter on a full production dataset. SHAP values would add per customer explanations that marketing teams need in order to defend individual targeting decisions.

Probability calibration. Tree ensemble raw probability outputs are miscalibrated by default. CalibratedClassifierCV would fix this at the cost of one more cross validation loop, which matters for the threshold versus profit analysis to hold up in production.

9. Conclusion

The project delivered an end to end binary classification pipeline for a European bank's term deposit marketing campaign, with a test ROC AUC of 0.740 and a 28 percent lift in expected profit over the call everyone baseline. The most consequential findings were operational rather than algorithmic: the bank is calling the wrong channel (telephone instead of cellular), in the wrong months (May instead of March, September, October, December), and treating all customers as if they had equal propensity when students, retired, and unemployed customers in fact convert at 2 to 3 times the baseline. The modeling pipeline is ready to run against production data; the specific numbers above are directional on the UCI sample and would tighten on current real data.